

MatMul — C = A × B on the T1 fabric (3×3 worked example)

Steps run top → bottom. Colour helps tracking here. The V-load transpose turns columns into B^T rows can be seen with colour; each Output cell keeps its final-position colour, so a result enters at column 0 and shifts right into place.

Example

0. Reference

C = A × B (true result)

Matrix A

123

456

789

Matrix B / B^T

B

101112

131415

161718

Intermediate

Output C

849096

201216231

318342366

Expected

Setup

1. Load H / V

vle8 v4,(A) [H]
csrwi 0x7c0,1
vle8 v5,(B) [V] → B^T

Matrix A

123

456

789

Matrix B / B^T

B^T

101316

111417

121518

Intermediate

Output C

Iteration 0 (Full Steps)

2. k=2 Broadcast

csrwi 0x7c0,1
vrgather.vx v6,v5,2 [V]

Matrix A

123

456

789

Matrix B / B^T

101316

111417

121518

Intermediate

121518

121518

121518

Output C

3. k=2 Multiply

csrwi 0x7c0,0
vmul.vv v6,v4,v6 [H]

Matrix A

123

456

789

Matrix B / B^T

101316

111417

121518

Intermediate

123054

4875108

84120162

Output C

4. k=2 Reduce

vredsum.vs v7,v6,v28
→ v7[0]=C[*][2]

Matrix A

123

456

789

Matrix B / B^T

101316

111417

121518

Intermediate

123054

4875108

84120162

Output C

96

231

366

5. Shift Right

vslideup.vi v8,v7,1
vmv.v.v v7,v8

Matrix A

123

456

789

Matrix B / B^T

101316

111417

121518

Intermediate

Output C

96

231

366

shift one column right →

SAME AS ABOVE 6. Broadcast, 7. k=1 Multiply

Iteration 1 (Simplified)

8. k=1 Reduce

vredsum.vs v7,v6,v28
→ v7[0]=C[*][1]

Matrix A

123

456

789

Matrix B / B^T

101316

111417

121518

Intermediate

112851

4470102

77112153

Output C

9096

216231

342366

9. Shift Right

vslideup.vi v8,v7,1
vmv.v.v v7,v8

Matrix A

123

456

789

Matrix B / B^T

101316

111417

121518

Intermediate

Output C

9096

216231

342366

shift one column right →

SAME AS ABOVE 10. Broadcast, 11. k=0 Multiply

Iteration 2 (Simp)

12. k=0 Reduce

vredsum.vs v7,v6,v28
→ full C
vse8 v7,(C)

Matrix A

123

456

789

Matrix B / B^T

101316

111417

121518

Intermediate

102648

406596

70104144

Output C

849096

201216231

318342366

Answer!!